ChatGPT

Interpretability in Computer Vision

Executive summary

The most important ten papers to read first, spanning the five categories in your scope, are: **CVPR** [1] **2017** *Network Dissection* for quantifying unit-level semantics; **ICCV 2017** *Grad-CAM* for practical class-discriminative attribution; **ICML** [2] **2018** *TCAV* for concept-level sensitivity analysis; **NeurIPS** [3] **2019** *ProtoPNet* for intrinsically interpretable prototype reasoning; **ICML 2020** *Concept Bottleneck Models* for human intervention on internal concepts; **CVPR** [4] **2021** *Transformer Interpretability Beyond Attention Visualization* for transformer-specific attribution; **Nature Machine Intelligence** [5] **2020** *Concept Whitening* for concept-aligned latent axes; **CVPR 2023** *CRAFT* for automated concept factorization and local/global concept explanations; **ICLR** [6] **2023** *What Do Self-Supervised Vision Transformers Learn?* for a probing-style account of how SSL objectives shape ViT representations; and **NeurIPS 2024** *SpLiCE* for sparse linear concept decompositions of CLIP [7] embeddings. Together, these papers define the modern map of vision interpretability: unit semantics, concepts, interventions, sparse bases, transformer-specific explanations, and rigorous evaluation. [8]

Two high-level conclusions emerge from the literature. First, the field has shifted from **pixel-space explanations alone** toward **concept-space and representation-space interpretability**, especially through prototypes, bottlenecks, concept factors, and sparse decompositions. Second, the strongest recent work is increasingly judged not just by visual plausibility, but by **faithfulness, intervention utility, and evaluation quality**, including human studies, causal tests, and careful attribution benchmarks. [9]

Citation counts below are approximate counts surfaced by indexed paper pages or search-result metadata on 2026-04-19 and should be treated as moving targets. [10]

Comparison tables

Selected recent papers

Category	Paper	Method type	Sup./Unsup.	Model scope	Key datasets/ tasks	Main contribution
Dictionary / sparse basis	**SpLiCE** (NeurIPS 2024)	sparse linear concept decomposition	unsupervised / post-hoc	CLIP / VLM	ImageNet-S, Waterbirds, Dollar Street	decomposes dense CLIP embeddings into sparse nonnegative concept mixtures

Category	Paper	Method type	Sup./Unsup.	Model scope	Key datasets/ tasks	Main contribution
Dictionary / concept factorization	**CRAFT** (CVPR 2023)	NMF concept factorization + Sobol importance	unsupervised concept discovery	CNNs	ILSVRC2012 / ImageNet, human utility tasks	links discovered concepts to both global and local explanations
Causal interventions	**VLG-CBM** (NeurIPS 2024)	grounded concept bottleneck	weakly supervised / hybrid	CLIP + grounded detectors	five standard image benchmarks	improves faithfulness and leakage control in concept bottlenecks
Causal interventions	**SCBM** (NeurIPS 2024)	stochastic concept bottleneck	supervised concepts	CNN / concept models	natural image benchmarks + synthetic settings	models concept dependencies so interventions propagate more realistically
Causal interventions	**LaBo** (CVPR 2023)	language-guided concept bottleneck	weakly supervised	CLIP + LM-guided concepts	11 image-classification datasets	removes the need for manually enumerating concepts and stays competitive
Probing	**MIS** (NeurIPS 2024)	automated per-unit interpretability metric	evaluation method	835 vision models	>70M units, ImageNet-scale explanations	makes large-scale interpretability measurement practical without humans

Category	Paper	Method type	Sup./Unsup.	Model scope	Key datasets/ tasks	Main contribution
Probing	**What Do Self-Supervised ViTs Learn?** (ICLR 2023)	representational probing / SSL analysis	unsupervised pretraining analysis	ViTs	ImageNet pretraining + downstream transfer	shows CL is more global/ shape-oriented and MIM more local/texture-oriented
Probing	**Analyzing ViTs in Class Embedding Space** (NeurIPS 2023)	class-embedding-space probing / mechanistic reverse engineering	post-hoc	ViTs	ImageNet-S	projects intermediate states into class space to track category building
Feature visualization	**Explaining in Style** (ICCV 2021)	GAN / StyleSpace counterfactual feature visualization	post-hoc	classifiers + StyleGAN	face / attribute-style image classifiers	uses identifiable latent directions to visualize and edit features causing predictions
Attribution	**Benchmarking Attribution Quality of Vision Models** (NeurIPS 2024)	evaluation benchmark	evaluation method	CNNs / ViTs / explainable models	standard vision backbones, 23 attribution methods	fixes key problems in deletion-style evaluation and compares model design choices
Attribution	**Transformer Interpretability Beyond Attention Visualization** (CVPR 2021)	relevance propagation for transformers	post-hoc	ViTs / transformers	ImageNet-style classification + transformer tests	establishes a principled transformer attribution pipeline beyond raw attention maps

Category	Paper	Method type	Sup./Unsup.	Model scope	Key datasets/ tasks	Main contribution
Attribution	**CRP** (Nature Machine Intelligence 2023)	concept relevance propagation	post-hoc	CNNs	ImageNet-scale concept analysis	unifies “where” and “what” by tracing concept-level relevance

Foundational papers

Category	Paper	Method type	Sup./Unsup.	Model scope	Key datasets/ tasks	Main contribution
Feature visualization	**Visualizing and Understanding ConvNets** (ECCV 2014)	deconvnet feature visualization + ablation	post-hoc	CNNs	ImageNet, Caltech-101, Caltech-256	established internal feature visualization as a debugging method
Feature visualization	**Feature Visualization** (Distill 2017)	activation maximization framework	post-hoc	CNNs	general framework	systematized regularization, optimization, and visualization objectives
Probing	**Network Dissection** (CVPR 2017)	unit-to-concept alignment	post-hoc	CNNs	Broden	made interpretability measurable at unit level
Causal / concept sensitivity	**TCAV** (ICML 2018)	concept activation vectors	supervised concepts	image classifiers + medical imagery	ImageNet-style tasks + medical example	tests directional sensitivity to human concepts
Dictionary / factorization	**Interpretable Basis Decomposition** (ECCV 2018)	concept basis decomposition	weakly supervised / post-hoc	CNNs	image classification explanations	decomposes predictions into interpretable basis components plus residual

Category	Paper	Method type	Sup./Unsup.	Model scope	Key datasets/ tasks	Main contribution
Dictionary / concept discovery	**DFF** (ECCV 2018)	NMF over activation maps	unsupervised	CNNs	object/part discovery on image datasets	discovers semantic parts by factorizing deep features
Dictionary / intrinsic interpretability	**ProtoPNet** (NeurIPS 2019)	prototype network	supervised	CNNs	CUB-200-2011, Stanford Cars	prediction by matching image parts to learned prototypes
Attribution	**Grad-CAM** (ICCV 2017)	gradient-weighted CAM	post-hoc	CNNs	classification, captioning, VQA	made saliency practical and class-discriminative
Attribution	**Integrated Gradients** (ICML 2017)	axiomatic path attribution	post-hoc	general differentiable nets	image, text, chemistry examples	introduced a principled axiomatic attribution baseline

Dictionary learning and sparse concept bases

A useful way to read this category is broadly: in the 2019–2024 venue slice you requested, the most influential "dictionary-like" work in vision is less about classical sparse autoencoders and more about **concept bases, factorization, whitening, and prototype dictionaries**. Pure vision SAE papers become much more visible after your cutoff; inside the cutoff, the papers below are the strongest matches to the underlying idea of decomposing internal representations into sparse, interpretable, reusable semantic atoms. This is a synthesis from the venue-constrained literature rather than a strict bibliometric claim. [48]

Usha Bhalla, Alex Oesterling, Suraj Srinivas, Flavio P. Calmon, and Himabindu Lakkaraju. "Interpreting CLIP with Sparse Linear Concept Embeddings." NeurIPS 2024. Approx. citations: **114**. Paper. [49] Code. [12] SpLiCE formulates interpretability for CLIP as sparse recovery: a dense multimodal embedding is decomposed into a sparse, nonnegative mixture of human-readable concepts. The important result is that the sparse concept representation is task-agnostic, usable without retraining, and still strong enough for downstream use cases such as bias diagnostics, spurious-correlation discovery, and even model editing. It matters because it is the clearest 2024 bridge between mechanistic sparsity ideas and computer-vision foundation models. Key experiments include ImageNet-S and robustness/bias case studies such as Waterbirds and Dollar Street. [50]

Thomas Fel, Agustin Picard, Louis Béthune, Thibaut Boissin, David Vigouroux, Julien Colin, Rémi Cadène, and Thomas Serre. "CRAFT: Concept Recursive Activation FacTorization for Explainability." CVPR 2023. Approx. citations: **237**. Paper. [51] Code. [14] CRAFT applies nonnegative matrix factorization to internal activations to discover high-level concepts and then estimates concept importance with Sobol

indices, connecting local and global explanations. The method stands out because it does not merely discover concepts: it also quantifies which concepts causally matter for a model's prediction behavior under perturbation-based sensitivity analysis. It matters for interpretability because it is one of the best recent concept-discovery pipelines that is both **automatic** and **faithfulness-aware**. Experiments center on ImageNet-scale models and human-centered utility evaluations. [52]

Zhi Chen, Yijie Bei, and Cynthia Rudin. "Concept Whitening for Interpretable Image Recognition." Nature Machine Intelligence 2020. Approx. citations surfaced by search: **549**. Paper. [53] Code. [54] Concept Whitening replaces ordinary whitening-style latent transforms with concept-aligned axes so that internal dimensions correspond to user-defined visual concepts. The method is important because it shifts interpretability from a purely post-hoc exercise to an architectural intervention: the latent coordinates themselves are trained to be semantically legible. It matters especially for researchers who want **concept-aligned internal spaces** rather than only after-the-fact saliency maps. The paper's key contribution is the alignment of hidden dimensions with user concepts while preserving competitive recognition accuracy. [55]

Chaofan Chen, Oscar Li, Chaofan Tao, Alina Jade Barnett, Jonathan Su, and Cynthia Rudin. "This Looks Like That: Deep Learning for Interpretable Image Recognition." NeurIPS 2019. Approx. citations: **about 2,000**. Paper. [56] Code. [44] ProtoPNet classifies by comparing image parts to learned prototypes and aggregating those prototype similarities into the final decision. The method is still one of the cleanest examples of **intrinsic visual interpretability** because the explanation and the computation share the same structure: the model literally reasons by "this looks like that." It matters because many later prototype, case-based, and concept-bottleneck papers inherit this design philosophy. Key experiments are on CUB-200-2011 and Stanford Cars. [57]

Bolei Zhou, Yiyou Sun, David Bau, and Antonio Torralba. "Interpretable Basis Decomposition for Visual Explanation." ECCV 2018. Approx. citations: **422**. Paper. [58] This paper decomposes a prediction into a small set of interpretable basis concepts plus an uninterpreted residual, rather than assuming a single saliency mask is enough. Its importance is methodological: it explicitly acknowledges that any human-readable explanation is partial and introduces a residual term to make that partiality visible instead of hiding it. That makes it foundational for later concept-basis and completeness-aware approaches. The paper evaluates explanation simplicity and faithfulness with human studies and image-classification experiments. [58]

Edo Collins, Radhakrishna Achanta, and Sabine Süsstrunk. "Deep Feature Factorization for Concept Discovery." ECCV 2018. Approx. citations: **136**. Paper. [59] Code. [42] DFF applies nonnegative matrix factorization directly to ReLU feature activations to reveal semantic parts and part hierarchies across images. It matters because it showed early that factorizing internal feature maps can expose meaningful visual parts without requiring part annotations, a line of thought that later feeds directly into CRAFT and related concept-mining work. Key experiments show semantic-part localization and hierarchical concept structure on object-centric image collections. [60]

Causal interventions

This is the category where vision interpretability has arguably become most **actionable**. Rather than asking only what information a model contains, these papers ask whether a human can **edit, correct, or intervene on internal concepts** and change downstream behavior in a predictable way. That shift — from explanation as observation to explanation as control — is one of the biggest advances since 2020. [61]

Divyansh Srivastava, Ge Yan, and Tsui-Wei Weng. "VLG-CBM: Training Concept Bottleneck Models with Vision-Language Guidance." NeurIPS 2024. Approx. citations: **66**. Paper. [62] Code. [16] VLG-CBM addresses a central weakness of recent automated CBMs: concept predictions can be unfaithful or leak label information. It introduces visually grounded concept supervision using grounded object detectors and proposes the Number of Effective Concepts to monitor leakage. It matters because it is one of the strongest recent attempts to make bottleneck concepts both **grounded** and **useful**, rather than merely convenient proxies. The paper evaluates on five standard image benchmarks and reports consistent gains over prior CBMs under faithfulness-aware metrics. [63]

Sonia Laguna, Ričards Marcinkevičs, Moritz Vandenhirtz, and Julia E. Vogt. "Beyond Concept Bottleneck Models: How to Make Black Boxes Intervenable?" NeurIPS 2024. Approx. citations: **40**. Paper. [64] Code. [65] This paper asks whether one can retrofit intervention capability onto already-trained black-box models given only a small concept-labeled validation set. The key result is that intervention need not require an intrinsically interpretable architecture from the start: intervention layers can be learned after the fact. It matters because it broadens the operational scope of concept interventions from "specialized interpretable models only" to "pretrained vision systems in general." [66]

Moritz Vandenhirtz, Sonia Laguna, Ričards Marcinkevičs, and Julia E. Vogt. "Stochastic Concept Bottleneck Models." NeurIPS 2024. Approx. citations: **47**. Paper. [67] Code. [18] SCBM relaxes the standard CBM assumption that concepts are independent and instead models concept uncertainty and dependencies jointly. This allows an intervention on one concept to propagate to correlated concepts, making human corrections more realistic and more effective. It matters because it upgrades CBMs from a convenient factorization to something closer to a proper latent-variable model of concept interaction. Key experiments include both synthetic settings and natural image benchmarks, including cases with CLIP-inferred concepts. [68]

Yue Yang, Artemis Panagopoulou, Shenghao Zhou, Daniel Jin, Chris Callison-Burch, and Mark Yatskar. "Language in a Bottle: Language Model Guided Concept Bottlenecks for Interpretable Image Classification." CVPR 2023. Approx. citations: **430**. Paper. [69] Code. [20] LaBo uses a language model to propose candidate concepts, then filters them with CLIP to construct a concept bottleneck without manual concept enumeration. The paper is important because it ties interpretability to the modern vision-language pipeline: concept spaces can now be partly discovered in language and grounded in images. It matters practically because it demonstrates competitive few-shot accuracy while keeping a bottleneck of human-readable concepts. The experiments span 11 diverse image-classification datasets. [70]

Tuomas Oikarinen, Subhro Das, Lam M. Nguyen, and Tsui-Wei Weng. "Label-Free Concept Bottleneck Models." ICLR 2023. Approx. citations: **386**. Paper. [71] Code. [72] LF-CBM removes the need for labeled concept data and scales concept bottlenecks to harder vision benchmarks, including ImageNet-scale settings. The core contribution is a practical pipeline that turns a high-performing black-box model into a concept bottleneck with minimal human effort. It matters because it substantially lowered the data-annotation barrier that had limited concept-bottleneck adoption in vision. The paper also includes a large-scale user evaluation of interpretability. [73]

Pang Wei Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. "Concept Bottleneck Models." ICML 2020. Approx. citations: **1,644**. Paper. [74] Code. [75] CBMs introduced the now-standard architecture that predicts human concepts first and uses those concepts for the final prediction, enabling test-time concept interventions. The paper is foundational

because it changed the field's target from "explain the black box" to "route decisions through editable, semantically meaningful variables." In vision, its flagship experiments are bird recognition on CUB and clinical prediction on osteoarthritis imaging data. [76]

Amirata Ghorbani, James Wexler, James Y. Zou, and Been Kim. "Towards Automatic Concept-based Explanations." NeurIPS 2019. Approx. citations: **1,110**. Paper. [77] Code. [78] ACE automates concept discovery by clustering image segments into candidate concepts and then evaluating their influence with TCAV. The paper matters because it removes the requirement that the analyst hand-specify every concept beforehand, making concept-level interpretation scalable. In practice, ACE became a key bridge between concept sensitivity analysis and unsupervised concept discovery in image classifiers. [79]

Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viégas, and Rory Sayres. "Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors." ICML 2018. Approx. citations: **3,163**. Paper. [80] Code. [39] TCAV measures how sensitive a classifier's prediction is to a human concept by estimating the directional derivative along a concept activation vector. It matters because it made concept-level explanation quantitative and falsifiable, rather than purely narrative. In vision, the zebra/stripes style examples became canonical because they show the jump from pixels to semantically interpretable factors. [81]

Probing

Probing papers ask what information is present in a representation, when it appears, and how that information evolves across layers or training objectives. In vision, the strongest recent work has focused on **transformers**, where probing is often a more reliable first tool than raw attention visualization. The literature increasingly blends classical linear probing with mechanistic-style projections into class or concept spaces. [82]

Roland S. Zimmermann, David Klindt, and Wieland Brendel. "Measuring Per-Unit Interpretability at Scale Without Humans." NeurIPS 2024. Approx. citations: **4**. Paper. [83] Code and interactive visualizations. [84] MIS automates the per-unit interpretability measurement problem by predicting human judgments from image-similarity structure, then validating the score with interventional psychophysics. The biggest result is scope: the method supports analysis over **more than 70 million units from 835 vision models**. It matters because interpretability research has historically been bottlenecked by tiny human studies; MIS turns interpretability into something that can be measured and optimized at scale. [21]

Michael A. Lepori, Alexa R. Tartaglini, Wai Keen Vong, Thomas Serre, Brenden M. Lake, and Ellie Pavlick. "Beyond the Doors of Perception: Vision Transformers Represent Relations Between Objects." NeurIPS 2024. Approx. citations: **16**. Paper. [85] Code. [86] This paper studies a same/different relational reasoning task and argues that pretrained ViTs often develop two stages of computation: a perceptual stage for object features and a relational stage for comparisons. It matters because it pushes probing in vision beyond texture/shape or category information toward **higher-order visual relations**, which are closer to algorithmic understanding. The work is especially relevant if your interest extends from probing to mechanistic interpretability. [87]

Martina G. Vilas, Timothy Schaumlöffel, and Gemma Roig. "Analyzing Vision Transformers for Image Classification in Class Embedding Space." NeurIPS 2023. Approx. citations: **45**. Paper. [88] Code. [25] The paper projects intermediate ViT representations into the learned class-embedding space and uses that

projection to track how image tokens become class-specific over layers. Its key finding is that image tokens progressively align with class prototypes and that attention and MLP blocks contribute differently to that alignment. It matters because it adapts the “logit lens” intuition from NLP to image classification in a way that is both lightweight and mechanistically suggestive. Key experiments use ImageNet-S. [89]

Namuk Park, Wonjae Kim, Byeongho Heo, Taekyung Kim, and Sangdoo Yun. “What Do Self-Supervised Vision Transformers Learn?” ICLR 2023. Approx. citations: **152**. Paper. [90] Code. [23] This paper compares contrastive learning and masked image modeling as pretraining objectives for self-supervised ViTs. The headline findings are memorable and useful: contrastive learning produces more global, shape-oriented, later-layer representations, while masked image modeling is more local, texture-oriented, and earlier-layer dominated. It matters because it links **training objective** to **representation content**, which is exactly the probing question most CV researchers care about. [91]

Maithra Raghu, Thomas Unterthiner, Simon Kornblith, Chiyuan Zhang, and Alexey Dosovitskiy. “Do Vision Transformers See Like Convolutional Neural Networks?” NeurIPS 2021. Approx. citations: **1,846**. Paper. [92] This paper is foundational for transformer probing in vision because it asks not merely whether ViTs work, but whether they learn the **same kinds of representations** as CNNs. It finds striking representational differences, including more uniform intermediate representations and earlier global aggregation in ViTs. It matters because later transformer-interpretability work, especially class-space probing and relation-focused studies, builds on this representational contrast. [92]

David Bau, Bolei Zhou, Aditya Khosla, Aude Oliva, and Antonio Torralba. “Network Dissection: Quantifying Interpretability of Deep Visual Representations.” CVPR 2017. Approx. citations: **2,263**. Paper. [93] Network Dissection scores hidden units by alignment with semantic concepts from the Broden dataset, turning “this unit looks interpretable” into a measurable quantity. It matters because it arguably created the modern vision-probing paradigm: identify a hidden unit, define a concept inventory, and quantify alignment layer by layer. Even researchers who have moved beyond single neurons still inherit its core evaluation instinct. [93]

Feature visualization

Feature visualization seeks to answer the question “what internal feature or unit is this model looking for?” The foundational literature here is exceptionally important, because many later concept, probing, and even attribution methods borrow its optimization machinery or its notion of human-legible internal features. In the recent 2019–2024 venue-constrained slice, the strongest papers tend to combine visualization with **counterfactual editing** or **shared-feature discovery**, rather than classical activation maximization alone. [94]

Alexander Lappe, Anna Bognár, Ghazaleh Ghamkhari Nejad, Albert Mukovskiy, Lucas Martini, Martin A. Giese, and Rufin Vogels. “Parallel Backpropagation for Shared-Feature Visualization.” NeurIPS 2024. Approx. citations: **5**. Paper. [95] This paper asks a subtle but important question: if two highly activating examples for a neuron are both strong, what *shared* feature is the neuron actually responding to? The method performs parallel backpropagation from paired examples to isolate common explanatory structure. It matters because feature visualization often overfits to a single optimized image; this paper instead targets feature **intersection**, which is often closer to semantic meaning. [95]

Oran Lang, Yossi Gandelsman, Michal Yarom, Yoav Wald, Gal Elidan, Avinatan Hassidim, William T. Freeman, Phillip Isola, Amir Globerson, Michal Irani, and Inbar Mosseri. "Explaining in Style: Training a GAN To Explain a Classifier in StyleSpace." ICCV 2021. Approx. citations: **213**. Paper. [26] Code. [27] Explaining in Style uses StyleGAN's StyleSpace as a semantically meaningful latent interface for explaining what a classifier relies on, and for generating visual edits that change the decision. Instead of optimizing pixels directly, it visualizes **editable latent causes**. It matters because it connects feature visualization to **counterfactual generation** and causal-style latent interventions, which made explanations both more semantic and more testable. [96]

Shan Carter, Zan Armstrong, Ludwig Schubert, Ian Johnson, and Chris Olah. "Activation Atlas." Distill 2019. Approx. citations: **254**. Paper. [97] Code. [98] Activation Atlas visualizes not one neuron at a time, but the geometry of many activations together, building explorable maps of what a network has learned. It matters because it helped the field move away from the idea that a single optimized image is enough to characterize a feature. In practice, it is a foundational bridge between feature visualization and representation analysis. [99]

Chris Olah, Alexander Mordvintsev, and Ludwig Schubert. "Feature Visualization." Distill 2017. Approx. citations: **1,579**. Paper. [100] Code infrastructure. [36] This article systematized the entire field of activation maximization: parameterizations, regularization, objectives, and the geometric interpretation of directions in activation space. It remains essential because almost every later feature-visualization paper, in vision or otherwise, inherits its design vocabulary. The article is also unusually pragmatic, which is why it still serves as a de facto methods manual. [101]

Jason Yosinski, Jeff Clune, Anh Nguyen, Thomas Fuchs, and Hod Lipson. "Understanding Neural Networks Through Deep Visualization." ICML Deep Learning Workshop 2015. Approx. citations: **about 2,600**. Paper. [102] This work introduced better regularized optimization-based visualizations and live activation tools for inspecting CNNs as they process inputs. It matters because it made clear that the raw optimization objective was not enough: regularization determines whether an optimized image becomes noise or a recognizable feature. Much of the later Distill-era work can be read as a refinement of this insight. [102]

Matthew D. Zeiler and Rob Fergus. "Visualizing and Understanding Convolutional Networks." ECCV 2014. Approx. citations: **about 26,000**. Paper. [103] Zeiler and Fergus introduced deconvnet-style visualizations and paired them with careful ablation studies to understand layer contributions. It matters because this is the paper that made internal visualization scientifically respectable in computer vision, not just visually intriguing. It also established the debugging ethos that still motivates a lot of interpretability work today. [103]

Attribution

Attribution remains the most widely deployed family of vision explanations, but the literature has become much more critical of naive saliency. The strongest recent papers are not merely new heatmap methods; they are papers on **transformer-specific propagation**, **evaluation protocol design**, and **clinical validation**, all motivated by the realization that visually appealing saliency can be misleading. [104]

Robin Hesse, Simone Schaub-Meyer, and Stefan Roth. "Benchmarking the Attribution Quality of Vision Models." NeurIPS 2024. Approx. citations: **7**. Paper. [105] Code. [29] This paper proposes an

improved evaluation protocol for attribution maps that addresses two long-standing issues in deletion-based evaluation: out-of-domain artifacts and poor cross-model comparability. Its importance is not the score itself, but the methodological correction: it shows that attribution quality depends materially on how evaluation is constructed. It matters because evaluation quality is now the bottleneck for making progress in attribution. The paper benchmarks 23 attribution methods across multiple model families and also studies how backbone design affects attribution quality. [106]

Adriel Saporta, Xiaotong Gui, Ashwin Agrawal, Anuj Pareek, Steven Truong, Chanh D. Nguyen, Van-Doan Ngo, Jayne Seekins, Francis G. Blankenberg, Andrew Ng, Matthew P. Lungren, and Pranav Rajpurkar. "Benchmarking Saliency Methods for Chest X-Ray Interpretation." Nature Machine Intelligence 2022. Approx. citations: **297**. Paper. [107] This paper is one of the most important cautionary studies in all of visual attribution. Across seven saliency methods and several architectures, the authors show that saliency maps can perform substantially worse than expert radiologist benchmarks for pathology localization. It matters because it forced the field to confront the gap between nice-looking heatmaps and clinically trustworthy explanations. The key experimental setting is chest X-ray interpretation on large clinical datasets. [107]

Hila Chefer, Shir Gur, and Lior Wolf. "Transformer Interpretability Beyond Attention Visualization." CVPR 2021. Approx. citations: **1,458**. Paper. [108] Code. [31] This paper derives a relevance propagation scheme through transformer blocks that handles attention, MLPs, and skip connections while preserving relevance conservation. It matters because it showed, convincingly, that **attention is not explanation**, and that transformers need their own attribution machinery rather than a direct reuse of CNN-era heuristics. For vision researchers, it became the standard entry point for transformer saliency beyond attention rollout. [109]

Suraj Srinivas and François Fleuret. "Full-Gradient Representation for Neural Network Visualization." NeurIPS 2019. Approx. citations: **447**. Paper. [110] Code. [111] FullGrad decomposes a prediction into input sensitivity and per-neuron sensitivity terms, aiming to satisfy completeness-style desiderata better than plain gradients. It matters because it reframes attribution as a representation involving both inputs and interior network biases, rather than only pixel gradients. This was an important conceptual step toward more faithful gradient-based explanations. [112]

Ruth Fong, Mandela Patrick, and Andrea Vedaldi. "Understanding Deep Networks via Extremal Perturbations and Smooth Masks." ICCV 2019. Approx. citations: **625**. Paper. [113] Code through TorchRay. [114] Extremal Perturbations puts perturbation-based attribution on firmer footing by optimizing smooth, area-controlled perturbations and tracing prediction changes as a function of mask size. It matters because perturbation methods are often more directly tied to model behavior than raw gradients, but historically were awkward and unstable; this paper made them practical and principled. It also extended perturbation analysis into intermediate layers, which is interpretability-relevant beyond input attribution alone. [115]

Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. "Sanity Checks for Saliency Maps." NeurIPS 2018. Approx. citations: **about 3,300**. Paper. [116] This paper delivered the most important negative result in attribution: several popular saliency maps change little when model weights are randomized, meaning they may reflect image priors more than learned model behavior. It matters because it permanently changed the standard for attribution evaluation and made "looks reasonable" an unacceptable validation strategy. Any serious attribution paper after 2018 has to reckon with these sanity checks. [116]

Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization." ICCV 2017. Approx. citations: **about 31,000**. Paper. [117] Grad-CAM maps the gradients of a target class into the final convolutional activations to produce coarse class-localization heatmaps. It matters because it became the default practical tool for vision attribution: model-agnostic across many CNN variants, cheap to compute, and easy to understand. Its influence is enormous even where later work has exposed its limitations. [117]

Mukund Sundararajan, Ankur Taly, and Qiqi Yan. "Axiomatic Attribution for Deep Networks." ICML 2017. Approx. citations: **about 10,700**. Paper. [118] Code. [47] Integrated Gradients introduced a path-based attribution satisfying sensitivity and implementation invariance, two axioms that became central in later debates about attribution correctness. It matters in computer vision because it offers a stronger theoretical baseline than raw gradients while remaining easy to implement. Many later comparisons still treat IG as a load-bearing reference method. [119]

Ruth C. Fong and Andrea Vedaldi. "Interpretable Explanations of Black Boxes by Meaningful Perturbation." ICCV 2017. Approx. citations: **about 2,300**. Paper. [120] Meaningful Perturbation formulates explanation as finding the smallest interpretable corruption that most affects the model's prediction. It matters because it made attribution testable in terms of explicit interventions on the input, rather than only backpropagated sensitivities. In the vision context, it set the stage for later perturbation-based methods such as Extremal Perturbations. [120]

Milestones and open problems

The timeline below captures the major milestones that shaped the field from feature visualization and saliency, through concept methods and bottlenecks, to sparse concept decompositions and large-scale evaluation. Each milestone is discussed in the sections above. [121]

```
timeline
    title Major milestones in computer-vision interpretability
    2014 : Visualizing and Understanding ConvNets
    2017 : Network Dissection
         : Grad-CAM
         : Feature Visualization
    2018 : TCAV
         : Deep Feature Factorization
         : Interpretable Basis Decomposition
    2019 : Activation Atlas
         : ProtoPNet
         : ACE
         : Extremal Perturbations
         : FullGrad
    2020 : Concept Whitening
         : Concept Bottleneck Models
    2021 : Transformer Interpretability Beyond Attention Visualization
         : Explaining in Style
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     : Do ViTs See Like CNNs
2022 : Benchmarking Saliency Methods for Chest X-Ray Interpretation
2023 : CRAFT
     : Label-Free CBM
     : Language in a Bottle
     : What Do Self-Supervised ViTs Learn
     : Analyzing ViTs in Class Embedding Space
2024 : SpLiCE
     : VLG-CBM
     : SCBM
     : Beyond CBM
     : Measuring Per-Unit Interpretability at Scale Without Humans
     : Benchmarking the Attribution Quality of Vision Models
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Several open problems stand out from this literature.

- **Vision-specific sparse autoencoders are still underdeveloped in top-tier venues through 2024.** Sparse concept decompositions are becoming important, but the clean "SAE for vision circuits" story is still less mature than the equivalent literature for language models. The natural next step is to connect sparse learned features to patch/token-level causal interventions in ViTs and VLMs. [122]
- **Concept faithfulness remains the central problem for concept bottlenecks.** Many recent papers are really about leakage, grounding, dependency structure, or intervention efficacy rather than about concept bottlenecks per se, which is a sign that the basic CBM idea is powerful but incomplete. Strong future work will probably combine grounded detectors, sparse concept dictionaries, and uncertainty-aware intervention policies. [123]
- **Transformer interpretability in vision is still earlier-stage than its NLP counterpart.** The best papers now probe class-space alignment, relation computation, SSL objective differences, and transformer-specific relevance propagation, but full circuit-level accounts of real image-classification behavior are still rare. The most promising direction is to combine class-space probing, sparse concept bases, and targeted internal perturbations. [124]
- **Evaluation remains a bottleneck.** Attribution research keeps circling back to benchmark design, sanity checks, or large-scale psychophysics replacements because the field still lacks universally trusted ground truth. Expect more work that treats evaluation itself as a first-class interpretability problem. [125]
- **Medical imaging has exposed the limits of plausible explanations.** Clinical benchmarking papers show that many saliency maps are not yet good enough for real diagnostic support, which means future medical interpretability work must prioritize validation, calibration, and task-specific human factors over prettier heatmaps. [126]
- **Cross-category unification is the most promising direction.** The best recent papers blur the old boundaries: CRAFT combines dictionary learning with attribution; CRP combines attribution with concepts; SpLiCE connects sparse decomposition, concept explanation, and monitoring; bottleneck work turns concept explanations into interventions. The next major breakthroughs will likely come from systems that jointly support **what**, **where**, and **what if** questions inside one framework. [127]

A final synthesis is worth stating plainly. For **input→output** explanations, the field's core lesson is that attribution alone is not enough unless it survives strong evaluation. For **representation→information** questions, probing and class-space analyses are especially powerful for ViTs. For **internal unit→meaning**,

feature visualization remains foundational but works best when coupled with datasets, concepts, or counterfactual editors. For **component→behavior**, concept bottlenecks are still the cleanest intervention interface in computer vision. And for **dictionary learning / SAEs**, the leading 2019–2024 vision papers point toward sparse concept bases, not yet a fully mature SAE canon — which is exactly why that direction looks so promising now. [128]

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